

Package Management in Linux Distributions: An Overview

Package management is a definitive feature of a Linux distro, as it offers power and speed in the installation of packages not offered by any graphical package management utility. This article will brief readers on package management in Linux distributions.



Applications in Linux distributions are called packages. A typical Linux distribution comprises a Linux kernel, GNU tools and libraries, additional software, documentation, a window system (the most common being the X Window system), a window manager, and a desktop environment.

There are almost 600 Linux distributions at present, of which close to 500 are in active development. A list of popular Linux distros can be found at www.distrowatch.com.

There are some old Linux distributions, from which new distributions have been forked. Examples include Debian, Red Hat, Arch, Gentoo, Slack, etc. You can get a complete tree at https://upload.wikimedia.org/wikipedia/commons/1/1b/Linux_Distribution_Timeline.svg.

Package manager	Used by
APK (Alpine)	Alpine Linux
APK (Android)	Android
Deb	Debian and its derivatives, such as Ubuntu, Xubuntu, and Linux Mint
ebuild	Gentoo Linux
eopkg	Solus
ORB	ORB Launcher
PISI	Pardus
PKG	OS X, iOS, PlayStation 3, Solaris, SunOS, UNIX System V, Symbian, BeOS, Apple Newton
.pkg.tar.zst	Arch Linux
PUP and PET	Puppy Linux (PUP format is deprecated since version 3.0)
RPM	Red Hat Enterprise Linux, Fedora; derivatives such as CentOS, and SUSE Linux Enterprise Server, openSUSE
SNAP	Compatible with most Linux distros as well as macOS, but mainly developed for Ubuntu

Table 1: A summary of package manager tools